

NETWORK SCIENCE · CITATION ANALYSIS · ACADEMIC INTEGRITY

Topological Signatures of Citation Cartels in Scale-Free Academic Networks

A Comprehensive Network Science Analysis

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OVERVIEW

Project Abstract & Motivation

- Analyzed a macroscopic citation network of **500,000 papers** to detect organized citation cartels
- Established mathematical baselines confirming **scale-free** and **small-world** properties
- Deployed a novel LLM pipeline for **Semantic Edge Annotation** to weight citations by true intent
- Flagged anomalous cliques using community detection, composite Z-scores, and Triad Motif profiling
- Simulated targeted attacks proving cartel eradication leaves global knowledge transmission **completely intact**

The Five Deliverables

1

Baseline Network Profiling

Structural decomposition of the citation graph

2

Scale-Free Verification

Mathematical proof via power-law fitting

3

Semantic Edge Annotation

LLM-weighted citation intent scoring

4

Community Detection & Cartel Scoring

Anomalous clique identification

5

Network Resilience Simulations

Targeted excision and impact analysis

Global Network Statistics

Modeled as a Directed Graph (DiGraph) where nodes represent papers and edges represent citations. The sampling generated a highly cohesive component with massive degree inequality and zero isolates.

Metric	Value
Total Nodes (N)	500,000
Total Edges (E)	4,871,544
Network Density	1.948×10^{-5}
Number of Isolates	0 (0.00%)
Average Degree	9.74
Max In-Degree	22,734

Null Models & Small-World Properties

Compared empirical data against Erdős-Rényi (ER) and Barabási-Albert (BA) null models.

- Empirical clustering ($C \approx 0.265$) proves exceptionally high local triadic closure
- Average path length ($L \approx 3.92$) confirms a dense **Small-World Web**

Metric	Empirical	ER Model	BA Model
Density	0.000019	0.000195	0.000360
Clustering Coef.	0.265034	0.000429	0.002532

Bow-Tie Topological Breakdown

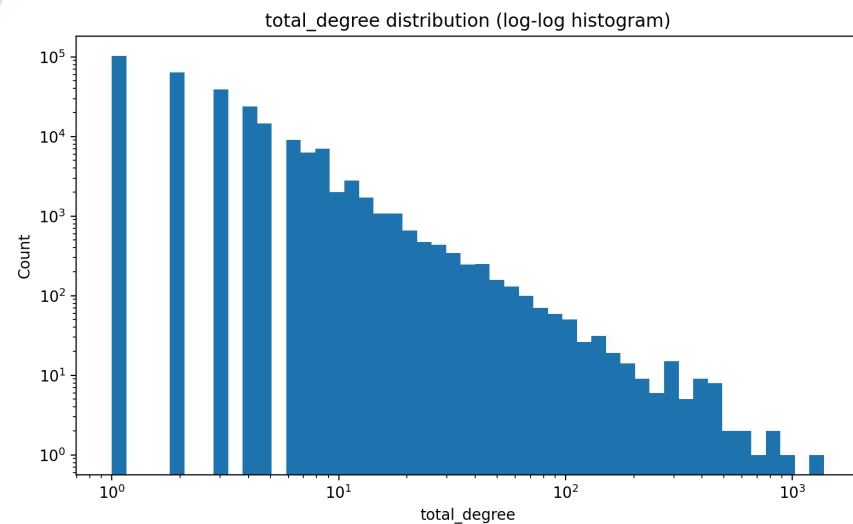
Mapped using Tarjan's algorithm. **78%** of the network exists in the IN-Component — tributaries citing the core. The global network remains highly **Disassortative**: mega-hubs are cited by peripheral papers, not by each other.

Component	Node Count	Fraction of Network
Giant SCC (Core)	12,872	2.57%
IN-Component	387,670	77.53%
OUT-Component	18,100	3.62%
Tendrils / Tubes	81,358	16.27%

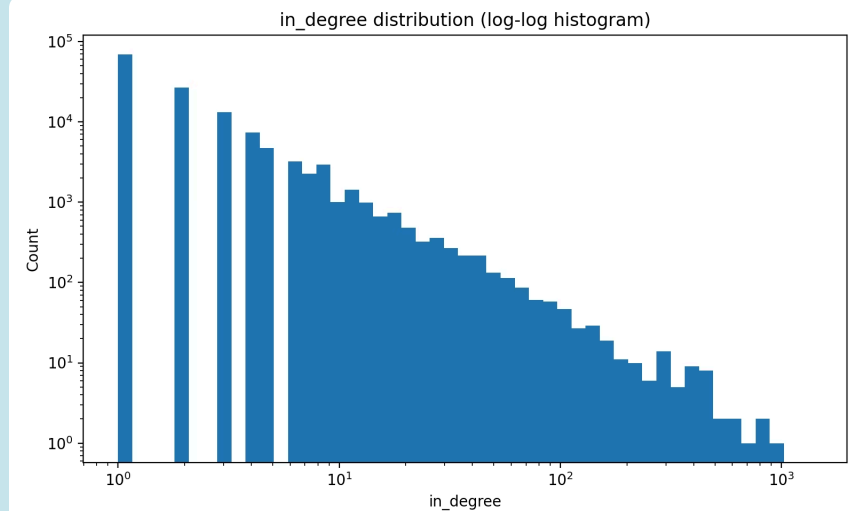
Analyzing Degree Distributions

Empirical distributions confirm that citation wealth is heavily centralized, aligning with **Pareto-like inequality curves**. Both in-degree and total-degree histograms exhibit the characteristic heavy-tailed signature of a scale-free network.

❏ **Left:** Figure: Log-Log In-Degree Histogram



❏ **Right:** Figure: Log-Log Total Degree Histogram



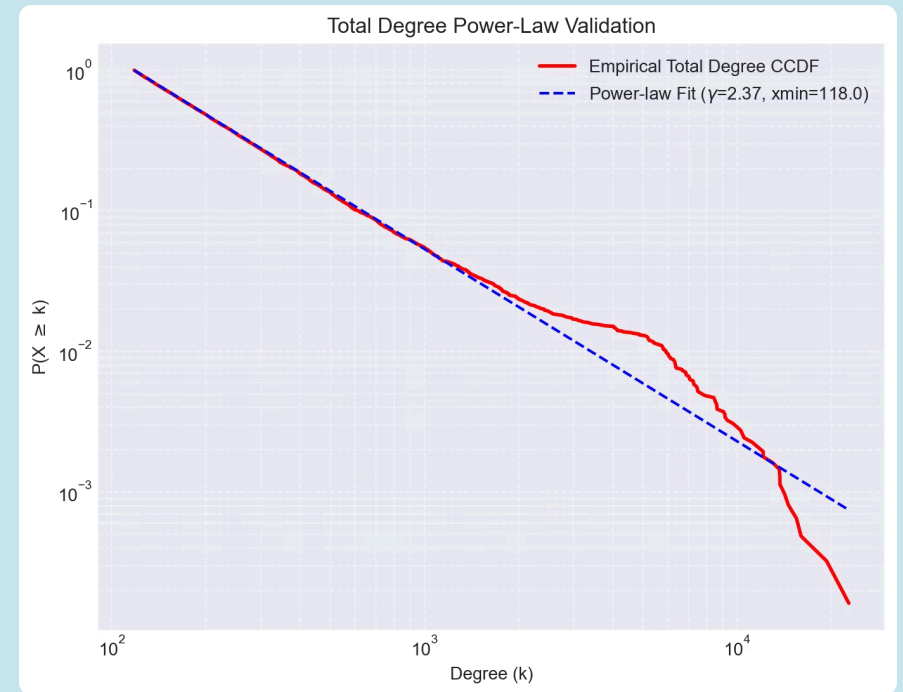
Maximum Likelihood Estimation (MLE)

Applied power-law MLE to fit the Complementary Cumulative Distribution Function (CCDF).

- **In-Degree** γ : 2.215
- **Out-Degree** γ : 2.956

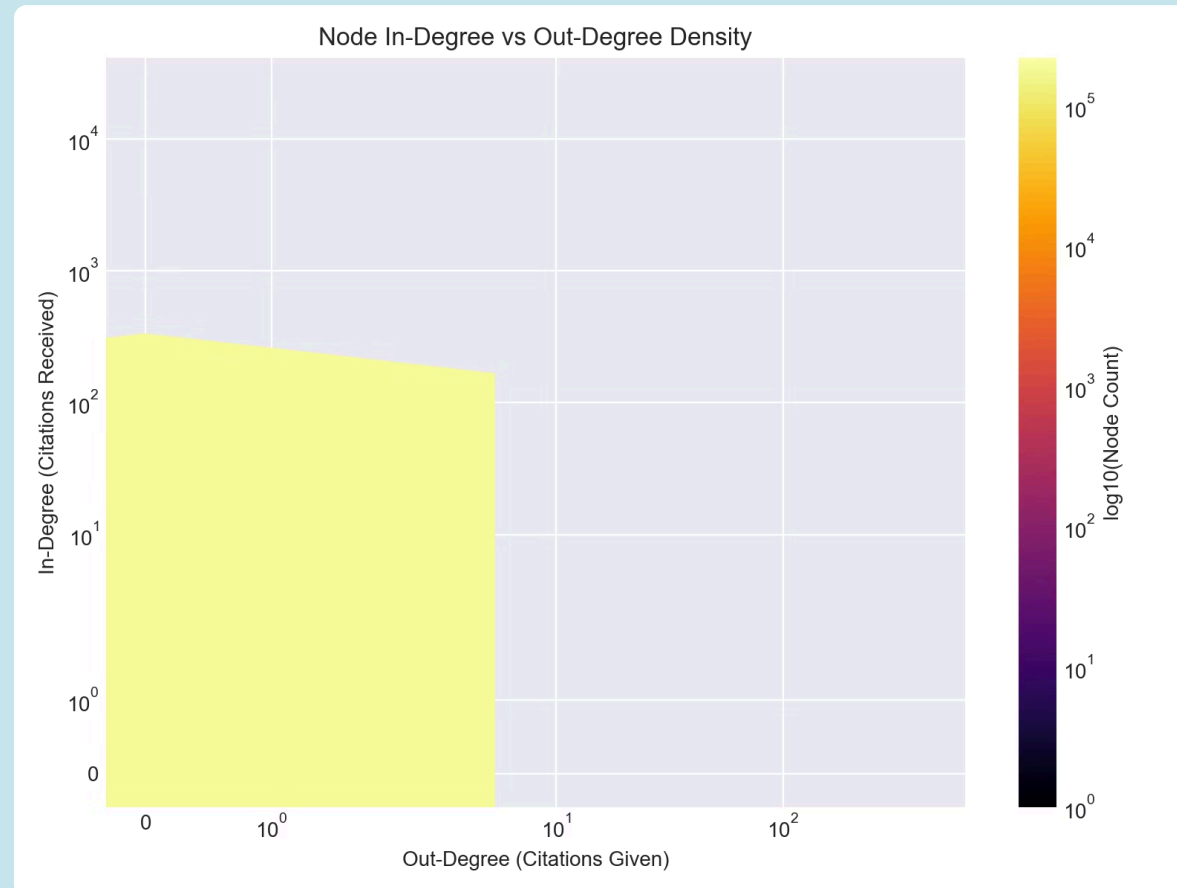
Because $2 < \gamma < 3$, this is mathematically proven as a **Scale-Free Network**. Growth is governed by **Preferential Attachment**.

Figure: Power-law exponential curve fits over empirical upper tails.



In-Degree Impact vs. Out-Degree Volume

Figure: Hexbin Density Plot tracking In-Degree impact versus Out-Degree volume.



Key Insight: Writing longer papers with more references (high out-degree) yields **mathematically zero relative benefit** to receiving citations (in-degree). Elite hubs reside exclusively along the Y-axis — citation impact is decoupled from reference volume.

Semantic Edge Annotation via LLMs

Standard graphs treat all citations equally. We weighted edges by true semantic intent using a Teacher-Student framework.

01

Teacher Labeling

Five concurrent Gemma models classified 3,000 abstract subsets to establish ground-truth citation intent labels.

02

Student Training

Fine-tuned SciBERT (FP16 precision) on the teacher-generated taxonomy for scalable semantic classification.

03

Graph Annotation

Converted the unweighted G into $G_{weighted}$ by scoring edges from "Perfunctory" to "Methodological".

Citation Intent Taxonomy & Semantic Weights

Each citation edge was assigned a semantic weight based on a fine-grained intent taxonomy, differentiating between substantive intellectual engagement and perfunctory referencing. This weighting provides a more nuanced understanding of influence within the network.

Intent Category	Semantic Weight	Cartel Relevance
Method	1.0	Highest impact; direct application or significant methodological reliance. Less susceptible to cartel manipulation due to technical specificity.
Result/Comparison	0.7	Substantial; directly comparing or building upon results. Cartels may attempt to inflate relevance through skewed comparisons.
Support	0.5	Moderate; provides evidence or general agreement. Can be used by cartels for soft endorsement and artificial bolstering.
Contrast/Criticism	0.3	Lower; critical engagement or contrasting findings. Unlikely to be used by cartels due to its adversarial nature.
Background	0.2	Minimal; general contextualization. Low-effort means for cartels to add references without deep engagement.
Perfunctory/Ceremonial	0.1	Lowest impact; token reference with no true intellectual contribution. Primary indicator of potential cartel behavior, used for artificial metric inflation.

Distribution of Ground-Truth Intents

Semantic weighting successfully detected instances where high-citation papers were artificially padded by superficial citations. The taxonomy below captures the distribution of citation intents across the corpus, revealing a pronounced imbalance between substantive and perfunctory referencing behavior.

Intent (High Depth)	Count
Method	354
Result / Comparison	436
Intent (Low Depth)	Count
Background	1,523
Perfunctory / Ceremonial	5

Key Findings: Weighted vs. Unweighted PageRank Shifts

Artificially Inflated Centrality

Papers like "Three-dimensional object recognition" and "Probabilistic object recognition" dropped thousands in rank due to superficial padding.

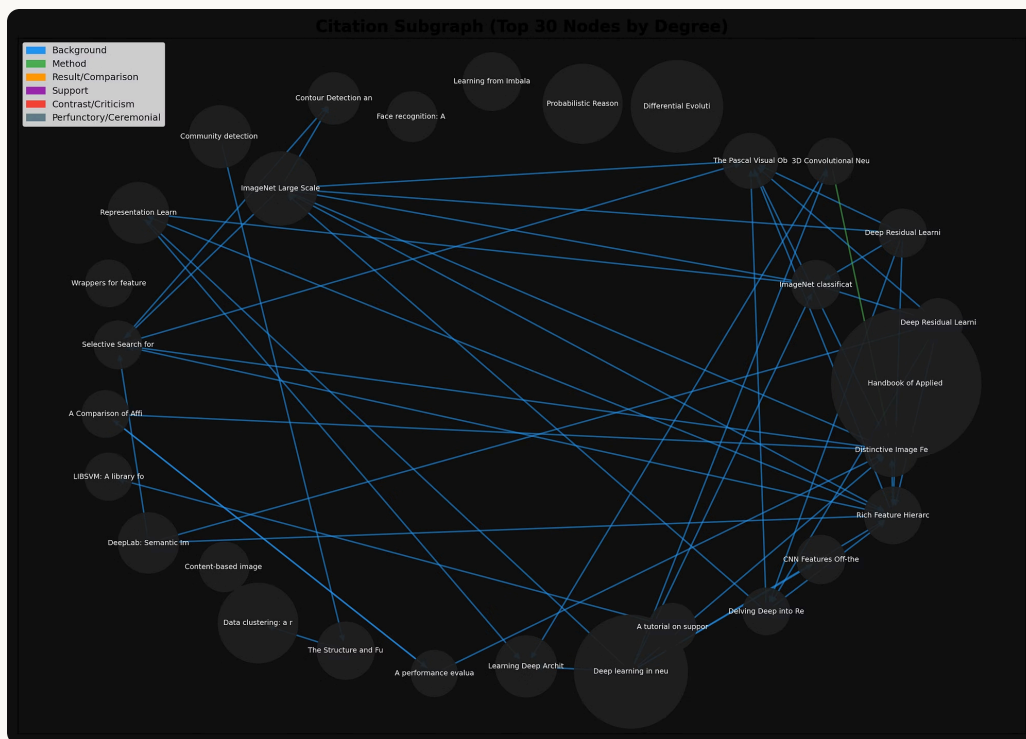
Genuine Deep Impact

Papers like "Mining Mid-level Features for Image Classification" and "Contour Tracking" jumped ranks due to high methodological citations.

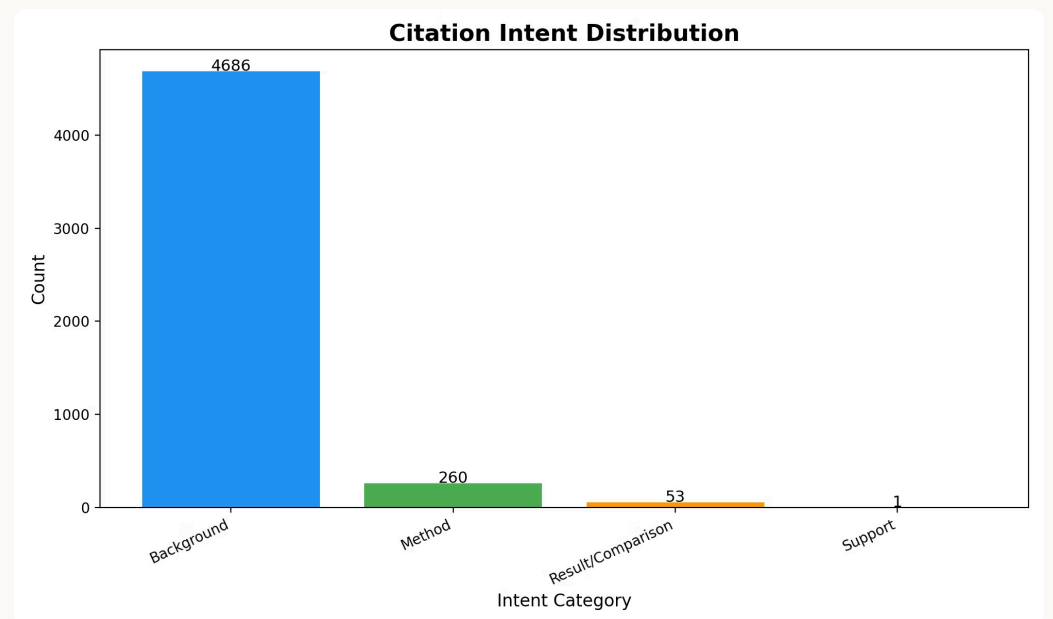
Identifying Artificially Inflated Centrality

Modulating centralities with semantic weights triggered massive rank shifts, exposing papers whose prominence depended on superficial padding rather than genuine contribution.

Annotated Semantic Sub-Graph



Intent Distribution



Dropped Ranks

Papers relying on superficial padding dropped **945+ positions** once low-depth citations were down-weighted.

- *Three-dimensional object recognition from single images* – dropped 945+ ranks
- *Probabilistic object recognition using multidimensional receptive field histograms* – dropped 945+ ranks
- *Object Recognition Using Multidimensional Receptive Field Histograms* – dropped 945+ ranks

Jumped Ranks

Papers with genuine methodological impact rose sharply, their true centrality previously obscured by noise.

- *Mining Mid-level Features for Image Classification* – jumped 1326 ranks
- *Contour Tracking by Stochastic Propagation of Belief* – jumped 1244 ranks
- *Efficient spatiotemporal grouping using the Nyquist sampling criterion* – jumped 1233 ranks

Community Superficiality Ratios

We introduced a **Superficiality Ratio** to quantify the prevalence of low-depth citations within specific communities (C) of papers:

$$\textit{Superficiality}(C) = \frac{|\text{Background} + \text{Perfunctory edges within } C|}{|\text{All edges within } C|}$$

Top-ranked communities often exhibited Superficiality Ratios approaching 1.0, reflecting a dominance of shallow citation intent. This indicated that their high centrality was largely propped up by perfunctory or background references rather than substantive intellectual engagement.

Communities with high internal citation density AND elevated Superficiality Ratios were subsequently flagged as potential citation cartel candidates for further investigation in Deliverable 4.

Isolating Suspicious Modules

The **Louvain Algorithm** isolated dense modules within the citation network, revealing 32 massive distinct communities with statistically significant structural cohesion.

32

Communities Detected

Distinct dense modules identified via
Louvain clustering

0.813

Observed Modularity (M)

Far exceeding the null random baseline

59.61

Statistical Z-Score

Guarantees communities are robust and
structurally non-random

❏ Null Random Modularity: $M_{rand} = 0.627 \pm 0.003$ — the astronomical Z-score of 59.61 confirms these communities are not artifacts of random graph structure.

Defining the Composite Cartel Index

The Z-Score Formula

Four structural signals are combined into a single composite metric quantifying cartel-like behavior:

Triad Census Analysis

Evaluated **16 directed MAN triad representations**. Cartels exhibit a highly specific **Triad Significance Profile (TSP)** — disproportionately heavy in direct cyclic loops and mutual triads, a structural fingerprint of coordinated manipulation.

01

Internal Citation Density

02

Null-Model Inflation
Ratio

03

Top-Down Reciprocity

04

Semantic Superficiality

Top Suspected Citation Cartels

Communities are prioritized by structural anomalousness. **Community 10** exhibits severe inflation and dense internal reciprocity – the strongest cartel signature in the network.

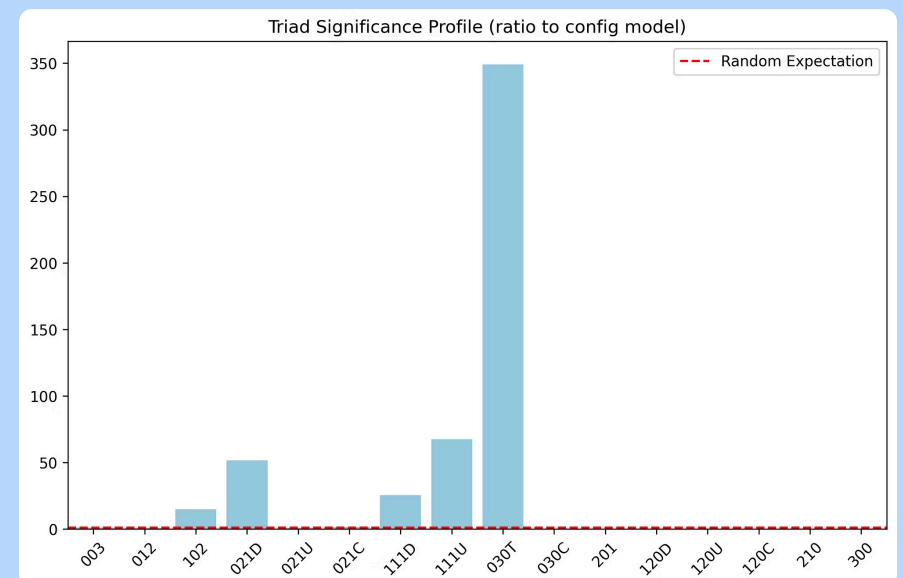
Community ID	Size (Nodes)	Cartel Index (Z)	Inflation	Reciprocity
10	11	1.235	49.63	0.14
30	31	0.771	2.00	0.00
25	26	0.539	3.00	0.00

Visualizing Cartel Topology & Triadic Signatures

- Left — Network Topology:** Force-directed layout isolating the cartel subgraph, revealing dense internal clustering and sparse external connectivity characteristic of coordinated citation rings.



- Right — Triad Significance Profile (TSP):** Manipulation signatures emerge clearly — cartels are over-represented in cyclic and mutual triad configurations, diverging sharply from random-network baselines.



Modeling Network Collapse

Tested how the Giant Component Size, $S(f)$, responds to different eradication strategies — probing whether cartels serve any genuine load-bearing scientific purpose.



Random Node Failure

Baseline robustness test — removing nodes uniformly at random to establish the network's natural tolerance.



Targeted Centrality Attacks

Strategic removal of degree-rich hubs and high-betweenness bridges to stress-test structural fragility.



Novel Edge Excision Experiment

Systematically deleting purely internal cartel edges to determine whether they transmit any global knowledge.

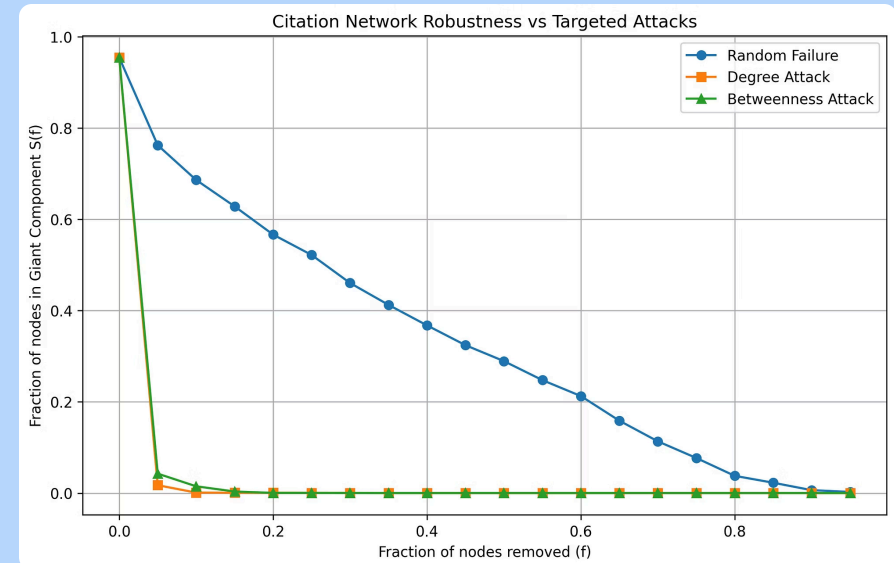
Random Failures vs. Targeted Attacks

As expected of scale-free networks, $S(f)$ dropped slowly during random node removal — the network absorbed diffuse failures with resilience.

However, targeting **degree-rich hubs** and **structural bridges** caused rapid collapse, matching behaviors modeled in the **Barabási-Albert** baseline. This confirms the network's scale-free fragility under adversarial attack.



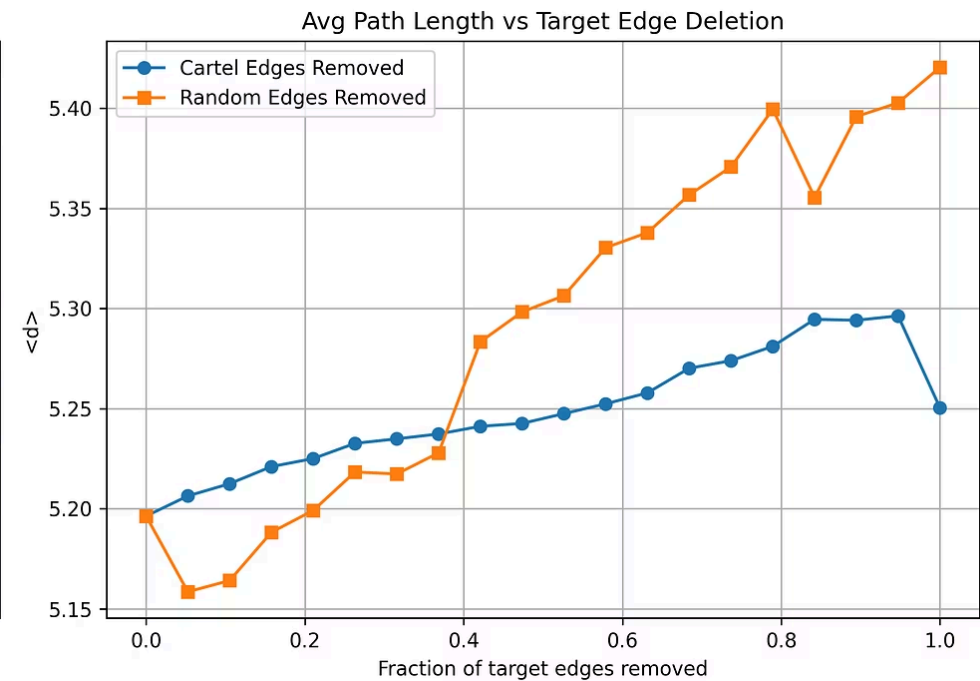
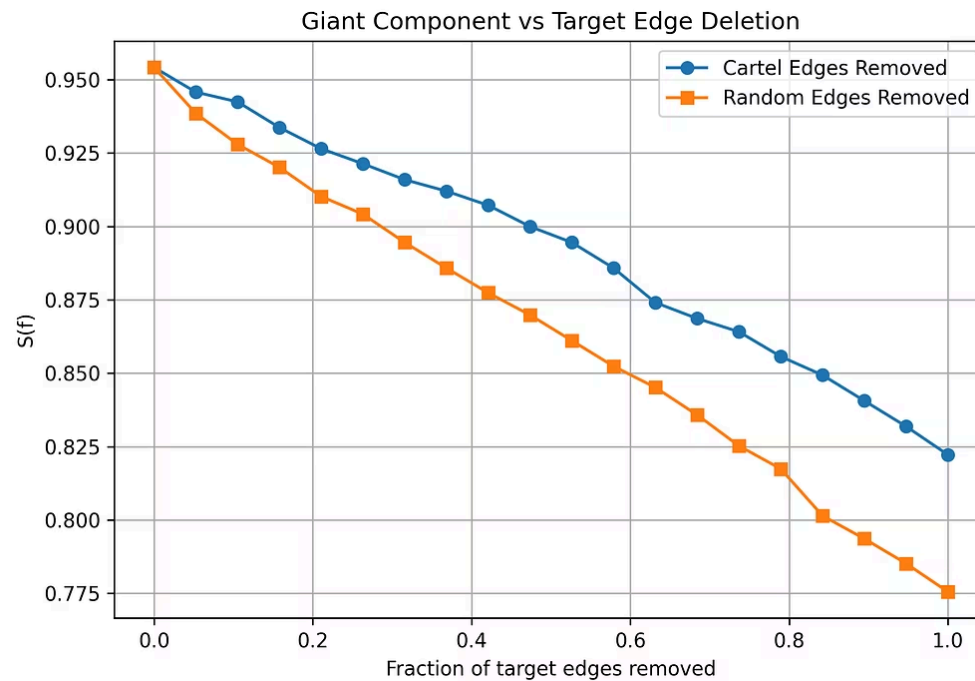
Figure: Network fragility chart — resilient gradual decay under random failure versus rapid targeted collapse when hubs and bridges are removed.



The True Nature of Cartel Edges

Eradicating the concentrated internal links forming Community 10 matched the **identical decay trajectory** of randomly clearing "junk" edges — compelling proof that cartels operate as superficial echo chambers with **zero global knowledge transmission value**.

Figure: $S(f)$ integrity sustains fully upon the complete removal of cartel operations — the network's scientific backbone remains entirely intact.



CONCLUSIONS

Final Conclusions

Robust Topological Signatures

Successfully documented structural fingerprints of organized citation fraud across 32 communities, validated by a Z-score of 59.61.

LLM Semantic Weighting Works

Demonstrated that LLM-based semantic weighting effectively unmask superficial citations, triggering massive PageRank corrections.

Mathematical Baselines Proven

Confirmed scale-free and small-world properties, establishing rigorous null-model benchmarks for anomaly detection.

Ultimate Finding

Academic networks are **impervious to the erasure of citation fraud** — deleting cartels resets artificial rank inflations while leaving genuine scientific methodology propagation perfectly intact.